RetailPulse | Executive Summary

Retail Sales & Customer Analytics - 2023 observation window

USC MS Business Analytics pre-program team case | Portfolio adaptation by Jenny Tsai

Recorded revenue (USD)	Transactions	Purchasers / registered	Return rate
166,909.56	**150**	**53 / 60**	**12.7%**

Findings and proposed investigations

Product concentration. Laptops contribute 34.8% of recorded revenue (58,101.12 USD) across 12 of 150 transactions. Review product-level returns and costs before drawing profitability conclusions.

East: review return records. East has 9 returns in 46 transactions (19.6%), versus 10 in the other 104 transactions (9.6%). Audit these nine records and compare product and payment mix before assigning a cause.

West: investigate volume and coverage. West records 26 transactions and the lowest regional revenue. Its average transaction value of 1,101.17 USD is close to the overall 1,112.73 USD; its return rate is below the overall rate. Check traffic, conversion and sampling coverage.

Customer prioritization. C051 has the highest heuristic RFM score (15). Revenue tiering and RFM describe different customer attributes; these scores do not predict churn or lifetime value.

Regional operating view

Region	*Revenue (USD)*	*Transactions*	*Avg. value (USD)*	*Returns*	*Return rate*
South	49,033.86	37	1,325.24	3	8.1%
East	47,602.70	46	1,034.84	9	19.6%
North	41,642.66	41	1,015.67	4	9.8%
West	28,630.34	26	1,101.17	3	11.5%
Total	**166,909.56**	**150**	**1,112.73**	**19**	**12.7%**

Data quality and decision limits

Five discounts and eight ratings are missing. Four revenue-formula exceptions, all involving missing discounts, total -261.77 USD under a 0% fill; source revenue is retained and unknown discounts are shown separately. Nine transactions predate loyalty signup and remain flagged. All 150 transactions match the 60-customer Week 5 master.

Revenue is after discount; refund treatment and costs are unknown. Rating comparisons use observed responses. Discount differences are observational, one year does not establish recurring seasonality, and sparse payment groups with repeated customers do not support a standalone significance claim. Recommendations are proposed investigations; no business impact is claimed.

github.com/chiatzujenny/retailpulse-analytics | Python, Pandas, NumPy, Matplotlib